

Setting Up and Deploying a Django Web Application with MongoDB on GCP

This article is a guide to setting up and deploying a Django Web application with MongoDB in GCP (Google Cloud Platform). Django is a high level Python framework which enables rapid development of applications with a clean, pragmatic design. MongoDB is a powerful database often associated with Web applications.



Let us get to work straight away. Begin by going through the following steps in sequence.

Creating a VM in GCP

Create an account or use your existing Google account to access *cloud.google.com*, then create a project and select 'VM Instance' in the 'Compute Engine' of the 'Compute' section.

Select 'Machine Type' based on your application requirement – I have selected 'Custom' (two vCPUs and 4GB memory). In the 'Boot disk' section, select OS and persistent disk size (I have selected 'Ubuntu 18.04 LTS' and 10GB standard persistent disk to start with).

Once the VM has been successfully created, you may note down the IP (your IP is ephemeral and for static IP,

you have to raise a request and there is a cost associated with it).

Enabling SSH access to your VM

In order to enable SSH access from your local machine, add your local RSA key to the VM in the section 'SSH Keys' of the metadata menu in 'Compute Engine'. After adding your RSA key, you can access the VM from your local machine using the following command:

e.g.: `ssh user_name@VM_IP`

To copy files to your new VM, you can use the `scp` command, as shown below:

e.g.: `scp /path/of/files user_name@VMIP:/path/of/remote/`

Installing Python

To check if Python 3 is already present, run the command given below:

```
python3 --version
```

If Python 3 is not present, install it using the commands given below:

```
sudo apt-get install python3.6
sudo apt-get install python3-setuptools
sudo apt-get update
```

Installing pip3

Check if pip3 is already installed by running the command given below:

```
pip3 --version
```

If it is not present, then install it by using the following command:

```
sudo apt install python3-pip
```